CRITICAL RANDOM DISCS COVER THE DISC

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ABSTRACT. Place discs of areas $\pi, \pi/2, \pi/3, \dots$ with independent centres uniform in the closed unit disc. A MathOverflow question asks whether they cover the unit disc almost surely. This is the critical case: with areas $c\pi/n$ the uncovered set has dimension $2(1-c)$, and at $c = 1$ known sufficient conditions do not decide it. We show that the answer is yes, assuming a recent exact criterion of Zhou for random ball coverings of tori, which has not yet been refereed. The interior reduces to Zhou's criterion by a measure-preserving localisation, and the criterion holds because two unit discs at distance s overlap in area at least $\pi(1-s)$. The boundary circle is a critical case of Shepp's theorem with random arc lengths and needs only classical tools. The analytic estimates are checked in Lean.

1. The question

Let $X_1, X_2, \dots$ be independent and uniform in the closed unit disc D, and let B_n be the open disc of radius $r_n = n^{-1/2}$, area π/n, centred at X_n. Is D covered almost surely? A MathOverflow user asked this [4]. The total area $\sum \pi/n$ is infinite, and a fixed point is covered infinitely often almost surely, since it lies in B_n with probability about $1/n$. The question is whether *every* point is covered.

The areas are exactly critical. With areas $c\pi/n$, the uncovered set of the Poisson version of the model has dimension $2(1-c)$ [1, Thm. 3.6], so the answer is no for $c < 1$ and yes for $c > 1$, and $c = 1$ is the boundary. At the boundary, lower-order terms decide. Biermé and Estrade give a necessary and a sufficient condition for Poisson balls in $\mathbb{R}^d$ and show that critical intensities can go either way [1, Props. 2.4, 2.5]; for this scale-invariant intensity their sufficient condition fails narrowly (the relevant expression tends to e^{-3} rather than to infinity), so they do not settle it.

Zhou has recently given an exact criterion for balls on a torus [6, Thm. 2.1]. Let $\mathbb{T}^d = (\mathbb{R}/\mathbb{Z})^d$ with Haar measure m, $d \geq 2$, let the centres be independent and uniform, the radii $r_1 \geq r_2 \geq \dots \downarrow 0$, and put $u_n(z) = m\big(B(0, r_n) \cap B(z, r_n)\big)$ and $H(z) = \sum_n u_n(z)$. Then every point is covered infinitely often almost surely if and only if $\int_{\mathbb{T}^d} e^H \, dm = \infty$.

Theorem 1. *Assume Zhou's criterion* [6, Thm. 2.1] *for $d = 2$. Then $D \subset \bigcup_n B_n$ almost surely.*

Only the interior of D uses Zhou's criterion; the boundary circle is covered almost surely by Shepp's theorem alone (Section 3).

2. The interior

Localisation. Fix p in the open disc and $\delta > 0$ with $B(p, 4\delta) \subset D$ and $4\delta < \sqrt{\pi}/2$. Let S be the square of side $\sqrt{\pi}$ centred at p, a fundamental domain of the torus $T = \mathbb{R}^2/\sqrt{\pi}\,\mathbb{Z}^2$, which has area π. Since $|D| = |S| = \pi$ there is a measure-preserving bijection $\varphi : D \to S$, up to null sets, that is the identity on $D \cap S$. Then $Y_n = \varphi(X_n)$ are independent and uniform on T. Put $K = \overline{B}(p, \delta)$. Once $r_n < \delta$, a torus ball $B(Y_n, r_n)$ that meets K has $Y_n \in B(p, 2\delta) \subset D \cap S$, so $Y_n = X_n$ and the torus ball is B_n. Hence if the balls $B(Y_n, r_n)$ cover every point of T infinitely often, then $K \subset \bigcup_n B_n$. Countably many such K cover the open disc.

The criterion. Scaling T to $(\mathbb{R}/\mathbb{Z})^2$ turns the radii into $\rho_n = (\pi n)^{-1/2}$, with $m(B(0, \rho_n)) = 1/n$, and

$$u_n(z) = \frac{1}{n}\,\psi\Big(\frac{|z|}{\rho_n}\Big),$$

where $\pi\psi(s)$ is the area of the lens in which two unit discs at distance s overlap. The lens contains the disc of radius $1 - s/2$ about the midpoint of the centres (Figure 1): a point within $1 - s/2$ of the midpoint is within $1 - s/2 + s/2 = 1$ of each centre. So

$$\psi(s) \;\geq\; \Big(1 - \frac{s}{2}\Big)^2 \;\geq\; 1 - s \qquad (0 \leq s \leq 2). \tag{1}$$

For small $t = |z|$ let $N = \lfloor 1/(\pi t^2) \rfloor$, so that $|z|/\rho_n = t\sqrt{\pi n} \leq 1$ for $n \leq N$. Dropping the terms with $n > N$, which are nonnegative, and using $\sum_{n \leq N} 1/n \geq \log(N+1)$ and $\sum_{n \leq N} n^{-1/2} \leq 2\sqrt{N}$,

$$H(z) \;\geq\; \sum_{n=1}^{N} \frac{1}{n}\big(1 - t\sqrt{\pi n}\big) \;\geq\; \log(N+1) - 2t\sqrt{\pi N} \;\geq\; 2\log\frac{1}{t} - \log\pi - 2.$$

So $e^{H(z)} \geq c\,|z|^{-2}$ near 0, and $\int e^H \, dm \geq c \int_0^{t_0} t^{-2}\, 2\pi t \, dt = \infty$. By Zhou's criterion the torus is covered infinitely often almost surely, and by the localisation the open disc is covered.

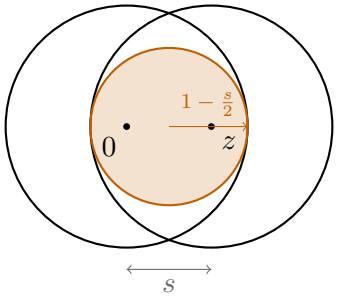


Figure 1. Two unit discs at distance s overlap in a lens that contains the disc of radius $1 - s/2$ about the midpoint, so the lens has area at least $\pi(1 - s/2)^2 \geq \pi(1 - s)$. This is the only estimate the interior needs.

3. The rim

For $|X_n| = \rho$, the set $B_n \cap S^1$ is empty if $\rho \leq 1 - r_n$ and otherwise an arc centred at the angle $\arg X_n$. Let ℓ_n be its length as a fraction of the circle. Since $\arg X_n$ is uniform and independent of $|X_n|$, conditionally on the lengths (ℓ_n) the arcs are independent with uniform positions. Shepp's theorem [5], [2, Ch. 11] then says that the circle is covered almost surely if and only if

$$\int_0^1 e^{F(t)}\, dt = \infty, \qquad F(t) = \sum_n (\ell_n - t)^+. \tag{2}$$

This is Shepp's series $\sum_N N^{-2} e^{\ell_1 + \cdots + \ell_N} = \infty$ for decreasing lengths, rewritten by summation by parts [6, §2, eq. (3)]; the integral form is symmetric in the ℓ_n, so their order and the empty arcs do not matter, and finitely many arcs change F by a bounded amount. We show that (2) holds almost surely.

The mean. By Fubini, $\mathbb{E}\,\ell_n = |B(q, r_n) \cap D|/\pi$ for any $q \in S^1$. The disc $B(q, r)$ contains the half-disc on the inner side of the tangent line at q, except for the sliver between that line and the circle, whose area is at most $\int_{-r}^{r} \big(1 - \sqrt{1 - x^2}\big)\, dx \leq \int_{-r}^{r} x^2\, dx = \frac{2}{3} r^3$. So $\mathbb{E}\,\ell_n \geq \frac{1}{2n} - \frac{2}{3\pi} n^{-3/2}$. Also $P(\ell_n > 0) = P(|X_n| > 1 - r_n) \leq 2r_n$, and $(\ell - t)^+ \geq \ell - t\,\mathbf{1}_{\ell > 0}$. Summing over $n \leq N = \lfloor t^{-2} \rfloor$,

$$\mathbb{E}\,F(t) \;\geq\; \tfrac{1}{2}\log N - C - 2t \sum_{n \leq N} n^{-1/2} \;\geq\; \log\tfrac{1}{t} - C'.$$

Fluctuations. An arc of chord at most $2r_n$ has $\ell_n \leq r_n/2$, so $\mathbb{E}\,\ell_n^2 \leq \frac{r_n^2}{4} \cdot 2r_n \leq n^{-3/2}$, which is summable; hence $G_0 = \sum_n (\ell_n - \mathbb{E}\ell_n)$ converges almost surely, by Kolmogorov's two-series

theorem. Since $(\ell - t)^+ = \ell - \min(\ell, t)$,

$$F(t) - \mathbb{E}F(t) - G_0 = -\sum_n \big(\min(\ell_n, t) - \mathbb{E}\min(\ell_n, t)\big),$$

whose variance is at most $\sum_n \mathbb{E}\min(\ell_n, t)^2 \le \sum_n 2r_n \min(t, r_n/2)^2 \le Ct$. With $t_k = 2^{-k}$, Chebyshev's inequality and the Borel-Cantelli lemma give $|F(t_k) - \mathbb{E}F(t_k) - G_0| \le 1$ for all large k, almost surely. Since F is nonincreasing,

$$\int_0^1 e^F\,dt \;\ge\; \sum_k 2^{-k-1} e^{F(t_k)} \;\ge\; \sum_{k \text{ large}} 2^{-k-1}\, 2^k e^{-C'-|G_0|-1} = \infty.$$

So, conditionally on the lengths, Shepp's criterion holds almost surely, and the circle is covered almost surely. Together with the interior this proves Theorem 1. □

4. Remarks

With areas $c\pi/n$ the same two computations give $H = 2c\log\frac{1}{t} + O(1)$ and $\mathbb{E}F(t) = c\log\frac{1}{t} + O(1)$, so both criteria hold exactly when $c \ge 1$: the interior and the rim become critical together, a consequence of the scale invariance of $r_n = n^{-1/2}$ in the plane. Numerically, at $c = 1$, $H(t) - 2\log\frac{1}{t}$ levels off near -0.42 and $\mathbb{E}F(t) - \log\frac{1}{t}$ near -2.72 for t from 10^{-1} to $10^{-3.5}$, while at $c = 0.9$ both drift away at the predicted rates.

The interior rests on [6], a single-author preprint of September 2026 that has not been refereed; before it, the case of Euclidean balls in the higher-dimensional Shepp problem was open (Kahane settled homothetic simplices; see the introduction of [6]). If Zhou's criterion stands, Theorem 1 is unconditional. The rim uses only Shepp's theorem.

5. Verification

The analytic estimates are formally verified in Lean 4 with Mathlib [3]: the lens containment and (1), the lower bound $\sum_{n\le N}\frac{1}{n}(1 - t\sqrt{\pi n}) \ge 2\log\frac{1}{t} - \log\pi - 2$ with the harmonic and square-root sums behind it, the divergence of $\int_0 t^{-1}\,dt$, and the sliver inequality $1 - \sqrt{1-x^2} \le x^2$. Zhou's criterion, Shepp's theorem, the localisation and the probabilistic estimates of the rim are not formalised. A script evaluates $H(t) - 2c\log\frac{1}{t}$ and $\mathbb{E}F(t) - c\log\frac{1}{t}$ numerically and checks that they level off for $c = 1$ and drift for $c = 0.9$. Code and Lean files are at [7].

Acknowledgments

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References

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